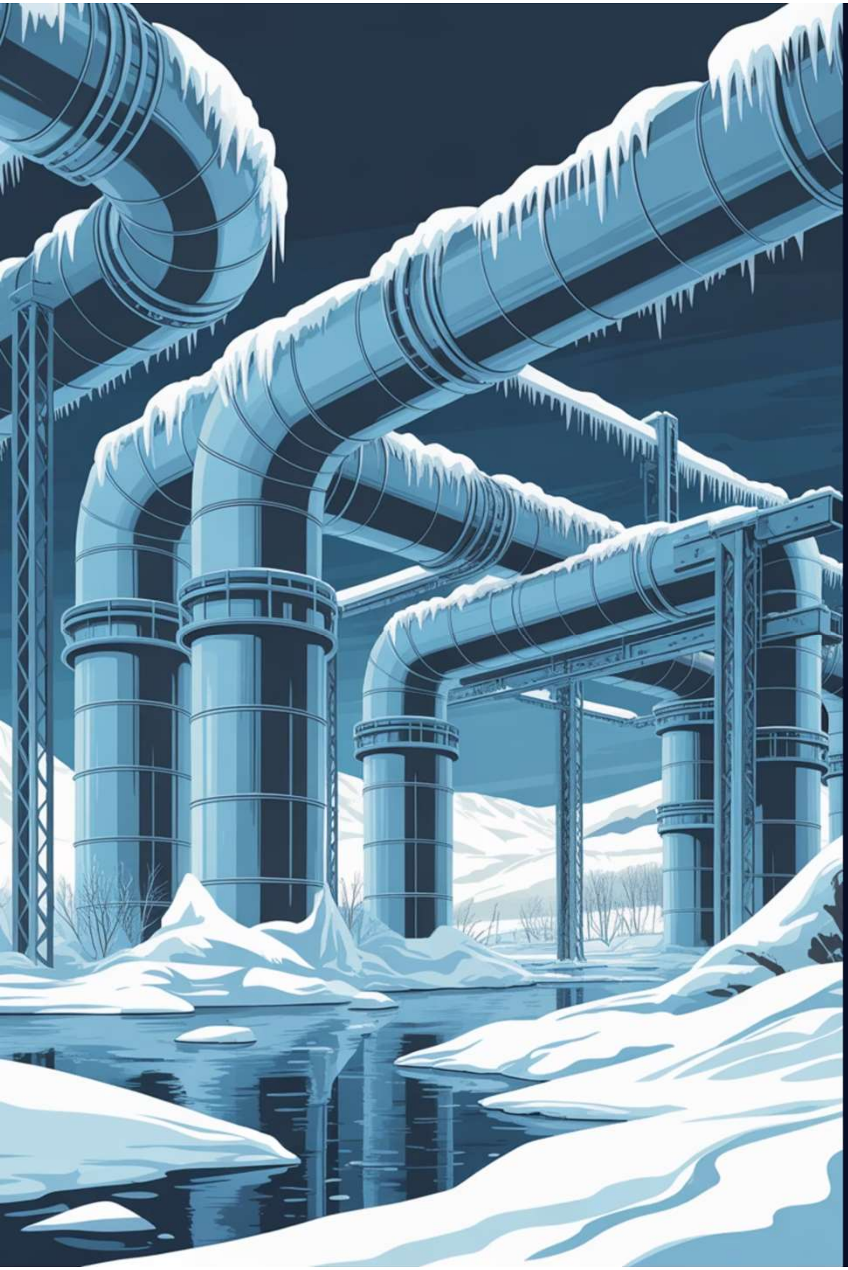
HEATSPOT OFF-GRID

Critical Off-Grid Thermal Protection

PIC-G21 · ElectroCap – 2026



CHAPTER 1.1 THE PROBLEM

Critical Infrastructure at Risk

Critical infrastructures in remote cold environments fail due to temperature.

Frozen Pipes

Ruptured tanks and burst pipes

Equipment Enclosure Freezing

Electronics and batteries lose performance

Telecom Outages

Remote connectivity becomes unreliable

Costly Repairs

Emergency maintenance expenses

> There is no simple, autonomous, low-power spot heating solution for off-grid assets.

CHAPTER 1.2 THE PROBLEM

Agricultural Yield at Risk

Cold snaps and unstable temperatures threaten greenhouse yields and beehive productivity.

Frost Damage

Night-time temperature drops can permanently reduce yield and quality

Energy Limitations

Grid power is unavailable or too expensive and requires fixed infrastructure

Localised Needs

Only critical zones need protection

Beekeeping Impact

Cold nights reduce colony performance and increase winter losses

> Agriculture and beekeeping need autonomous, low-power spot heating for critical protection off-grid.

CHAPTER 1.3 THE PROBLEM

BESS Performance at Risk

Cold temperatures reduce battery performance and can trigger failures in off-grid energy storage systems.

Capacity Drop

Reduced autonomy and unexpected shutdowns when energy is most needed.

Power Delivery Limits

Higher internal resistance impacts inverter performance and load support.

Protection Events and Downtime

BMS may restrict operation at low temperature.

Costly Field Maintenance

Downtime, travel costs, and component stress increase total cost of ownership.

> Off-grid BESS need targeted thermal protection to perform consistently, fully independent of the grid.

CHAPTER 4 VALUE PROPOSITION

Why We Are Different

Ultra Low Power
30-50W consumption

Localized Heating
Spot heating, not full enclosure

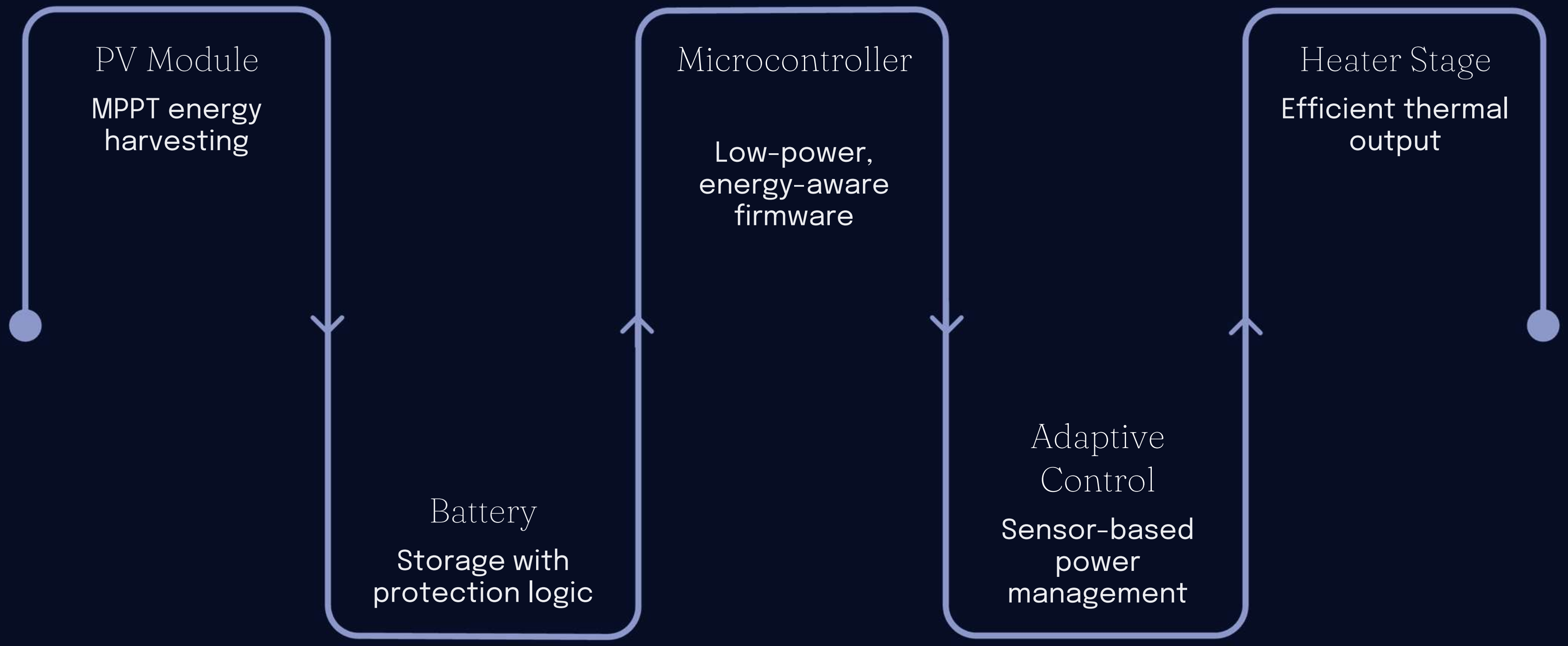
Multi-Day Autonomy
Solar-powered resilience

Modular & Scalable
Low install & maintenance cost

> Resilience without grid dependency.

Technology Architecture

Integrated system with energy-aware firmware for maximum autonomy.



Each module communicates through an energy-aware firmware layer that dynamically balances harvesting, storage, and consumption.

Competitive Landscape

Alternative	Autonomous	Intelligent
Manual (blankets, gas)	❌	❌
Grid-connected industrial	❌	Partial
DIY battery heaters	Partial	❌
Niche solar beehive systems	✅	❌
PIC-G21 (Ours)	✅	✅

Innovation gap: No modular, intelligent, low-power, asset-focused solution exists on the market today.

CHAPTER 7 CHALLENGES

Technical Challenges

PV & Battery Sizing

Worst-case winter scenarios

Low-Temp Battery Mgmt

Safe operation below 0°C

Heater Drive Electronics

Efficient power delivery

Thermal Control Algorithms

Stable, adaptive regulation

Ultra-Low-Power Firmware

Maximized energy autonomy

We address these with **integrated energy modelling** and **robust embedded control**.

CHAPTER 8 VALIDATION

Validation Metrics

Data-driven validation, not qualitative claims.

Multi-Day Autonomy

Operation without sunlight

Temp Accuracy

Maintain target temperature within ±2 °C

Energy Ratio

Harvesting vs. consumption balance

Fault Tolerance

Safe-state transitions

Comms Reliability

If remote monitoring enabled

CHAPTER 9 ROADMAP

Execution Plan

Structured development roadmap with an **iterative build-test-improve cycle**.

1. Requirements
   Definition & scoping
2. Hardware
   Design & assembly
3. Firmware
   Embedded development
4. Module Testing
   Component-level validation
5. System Validation
   Integrated testing
6. ElectroDay
   Final prototype demo